

SmartCity Healthcare Network Design

You have been assigned the responsibility of designing the network infrastructure for SmartCity Healthcare, a modern healthcare system connecting multiple medical units across a metropolitan city.

The network will support hospitals, clinics, research centers, and emergency units to ensure real-time patient data sharing, secure communication, and high availability.

The number of devices in each unit is given below:

- Main Hospital (MH) → 300 Devices
- Diagnostic Center (DC) → 150 Devices
- Emergency Response Unit (ERU) → 80 Devices
- Research Institute (RI) → 120 Devices
- Pharmacy Hub (PH) → 100 Devices
- Remote Health Clinic (RHC) → 60 Devices

Requirements

Topology Design & Server Setup

- Each medical unit will be represented by a **router**.
- The **Main Hospital (MH)** acts as the **central backbone** of the network.
- MH is directly connected to: Research Institute (RI), Emergency Response Unit (ERU)
- Diagnostic Center (DC), Pharmacy Hub (PH), and Remote Health Clinic (RHC) are interconnected through a **switch**, which then connects to MH.
- Emergency Response Unit (ERU) must maintain **dual connectivity**:
 - One link to MH
 - One link to RI (for redundancy)
- Each unit must have its own **Email Server** configured for internal communication:
 - Example format: mail.mh.local, mail.ri.local, etc.
- The **Main Hospital (MH)** hosts:
 - A **DNS Server**
 - A **Web Server** with domain: www.smartcityhealth.local
- Proper configuration must ensure that:
 - All devices can access the web server
 - Email communication is successfully established between MH and RI

DHCP Configuration

- Diagnostic Center (DC) and Emergency Response Unit (ERU):
 - Their routers will act as **DHCP servers**
- Pharmacy Hub (PH) and Remote Health Clinic (RHC):
 - Will use **dedicated DHCP servers**
- Main Hospital (MH) and Research Institute (RI):
 - All devices will use **static IP addressing**

Addressing

- Generate the base network using your group's Student ID:
 - **2 members:** Last 2 digits of second member's ID
 - **3 members:** Middle 2 digits of third member's ID
 - **4 members:** Sum of last 3 digits of fourth member's ID
- This value will form the **first two octets** of the network.
- The remaining octets will be 0.0 with a **/16 subnet mask**.

Example (4 members):

Student ID: 20219876

Last 3 digits: 8, 7, 6 → Sum = 21

Network Address → **21.21.0.0/16**

- Apply **VLSM (Variable Length Subnet Masking)** to divide the network according to device requirements of each unit.

Routing

- Different routing techniques must be implemented:
 1. **Static Routing:**
 - MH → DC: Next-hop static routing
 - ERU → MH: Exit-interface static routing
 - PH → DC: Recursive static routing via RI
 - RHC → MH: Floating static route via ERU (backup path)
 2. **Dynamic Routing:**
 - RI, PH, DC, and ERU will use **RIPv2** for routing updates
 - MH will participate in both:
 - Static routing
 - RIPv2 dynamic routing

Special Conditions

- If the link between **MH and RI fails**:
 - Traffic must automatically reroute through **ERU using RIPv2**
- Remote Health Clinic (RHC):
 - Must always maintain:
 - Primary path → via PH (RIPv2)
 - Backup path → via ERU (floating static route)
- Pharmacy Hub (PH):
 - If the primary path fails, it must use the **recursive static route via RI**

Deliverables

You must complete and submit the following:

- A fully configured **Cisco Packet Tracer (.pkt / .pka)** file
- A clearly labeled **network topology diagram**
- A detailed **IP Addressing Table** including:
 - Routers
 - PCs
 - Servers
- A document (PDF) containing:
 - VLSM subnetting design
 - Router configuration commands
 - DHCP configuration
 - DNS and Web server setup
 - Email server configuration (MH ↔ RI communication)
 - Routing configurations:
 - Static routes
 - RIPv2 setup
 - Floating static routes
 - Recursive routing
- **Work Distribution:**
 - Clearly mention the contributions of each group member